Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.2579703503345398, median 0.24113562030380015, std: 0.13942165236625975
Gyroscope error (imu0): mean 0.23130751933232505, median 0.08064351494537916, std: 0.6096335076086529
Accelerometer error (imu0): mean 0.5809624404836834, median 0.48935656072944245, std: 0.5907753405391876

Residuals

Reprojection error (cam0) [px]: mean 0.2579703503345398, median 0.24113562030380015, std: 0.13942165236625975
Gyroscope error (imu0) [rad/s]: mean 0.009132979851962906, median 0.003184140313784818, std: 0.024070858388619665
Accelerometer error (imu0) [m/s^2]: mean 0.22555746213260547, median 0.18999166938258083, std: 0.2293673002192771

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.10537367 -0.98691304 0.12206162 -0.00148523]
 [0.17856851 -0.1395285 -0.97398413 0.00302424]
 [0.97826871 -0.08083592 0.19093422 -0.00994051]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.10537367 0.17856851 0.97826871 0.00902796]
 [-0.98691304 -0.1395285 -0.08083592 -0.00184738]
 [0.12206162 -0.97398413 0.19093422 0.00502484]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.023207875653042633

Gravity vector in target coords: [m/s^2]
[-0.47697939 -9.79404305 -0.13279419]

Calibration configuration
==========

cam0

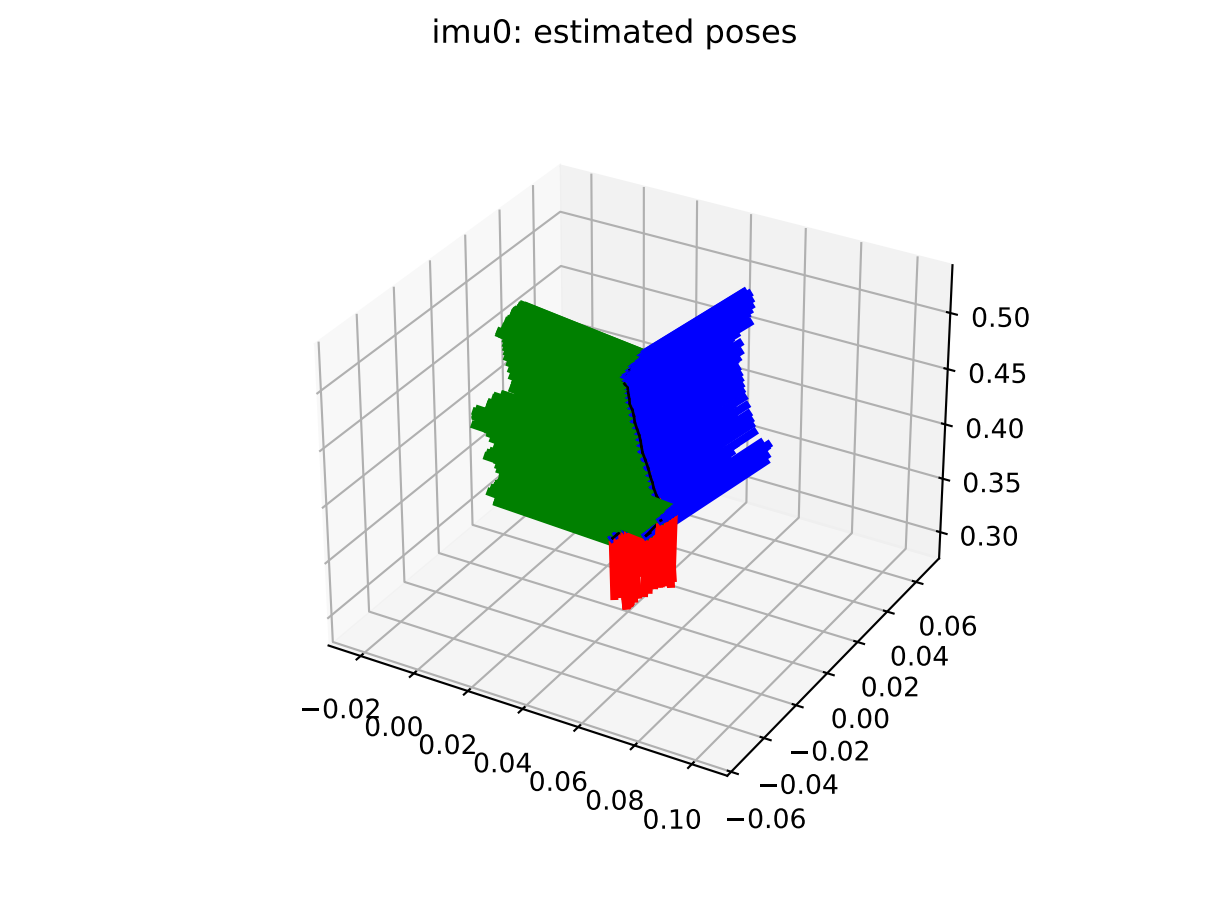
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

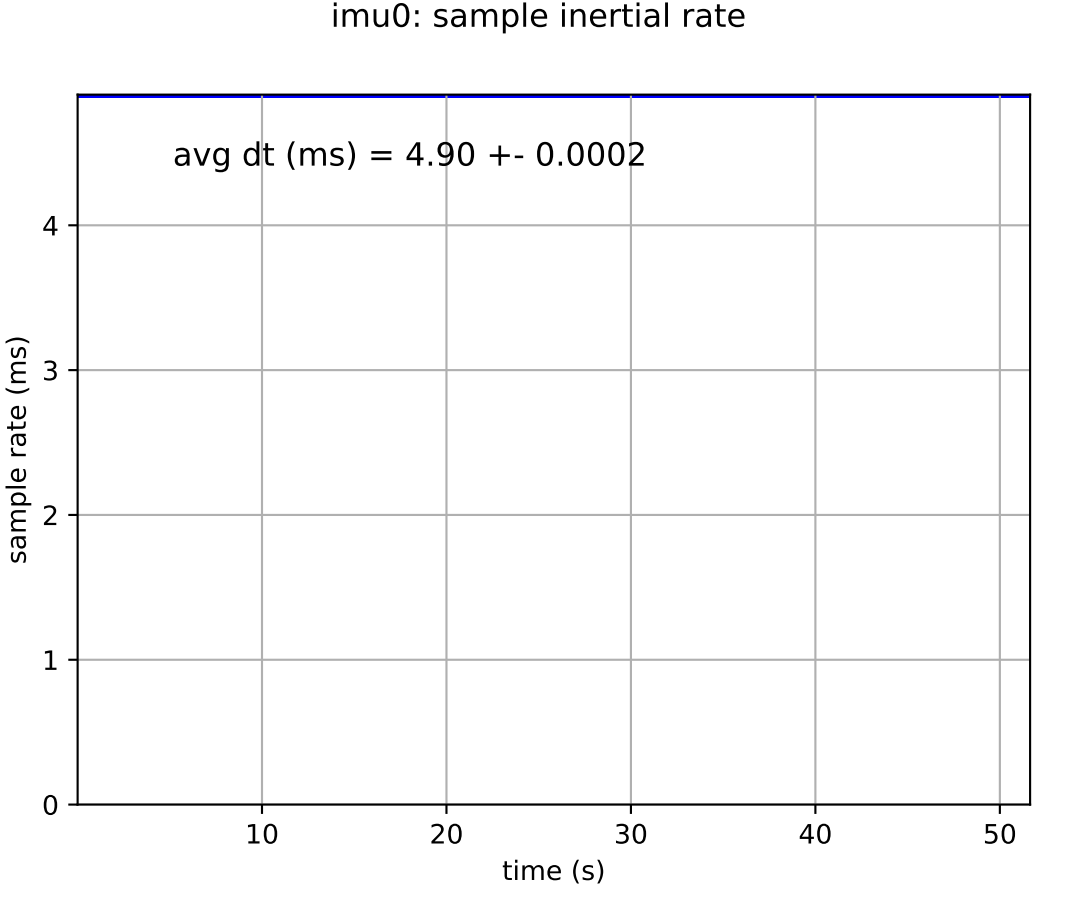
IMU configuration
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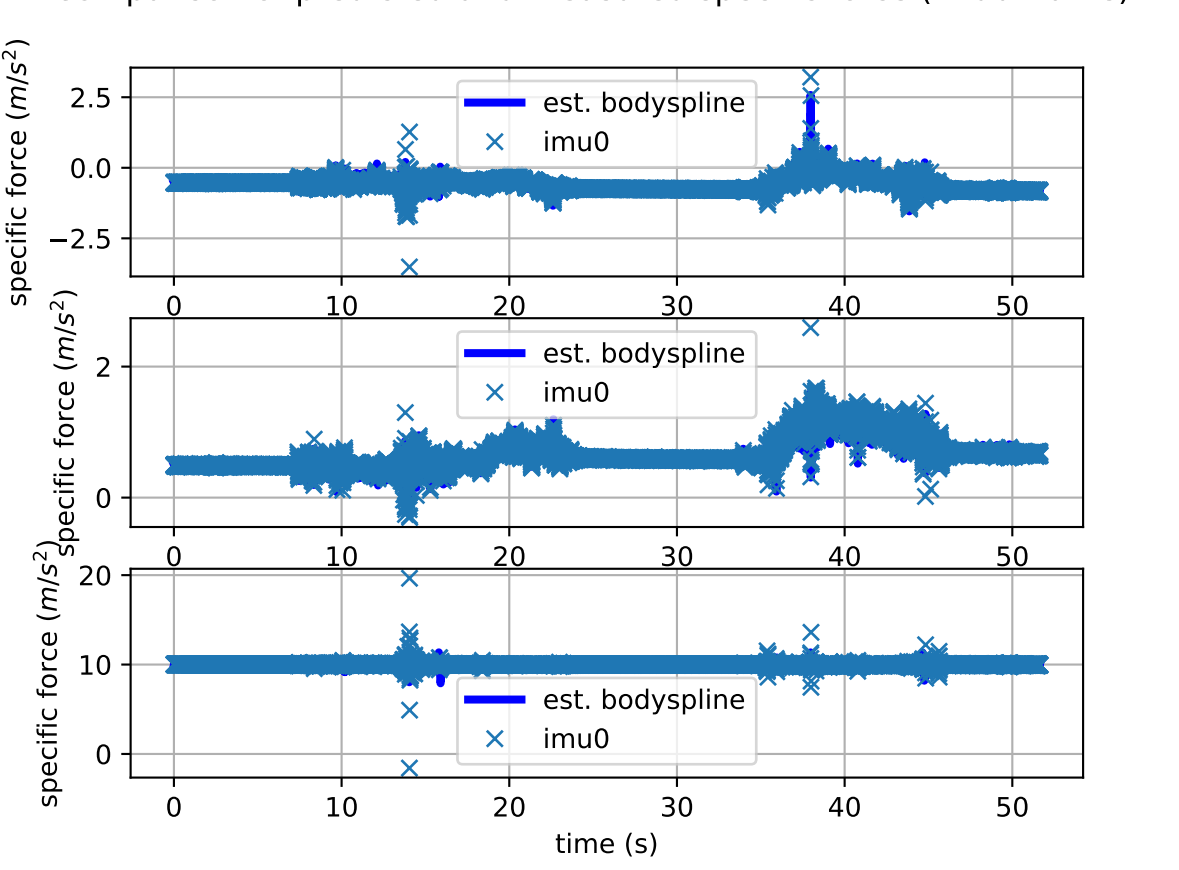
IMU0:

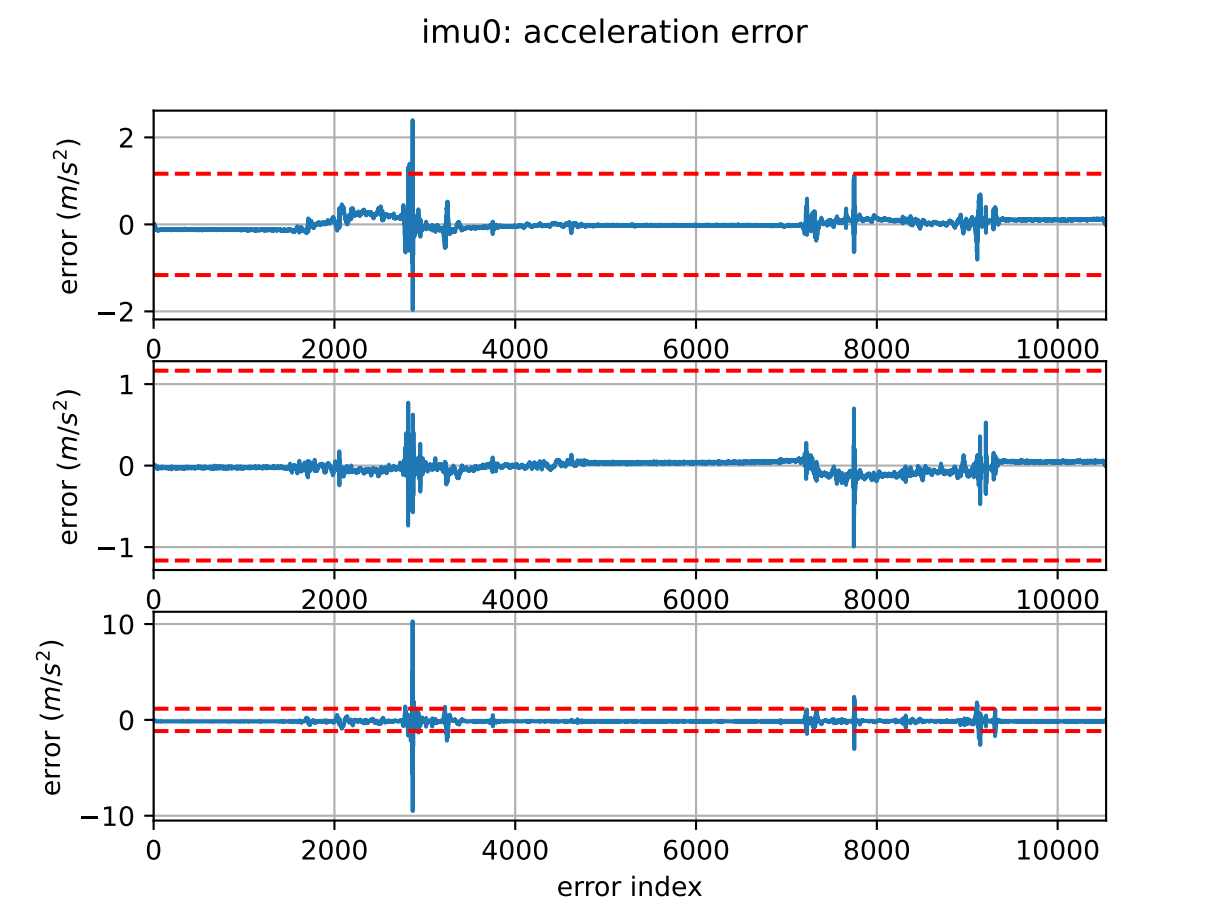
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.027176766288825538
    Noise density (discrete): 0.38824792519257595
    Random walk: 0.0067023621585376265
  Gyroscope:
    Noise density: 0.002763830369432253
    Noise density (discrete): 0.039484145947029654
    Random walk: 0.00014770260860981916

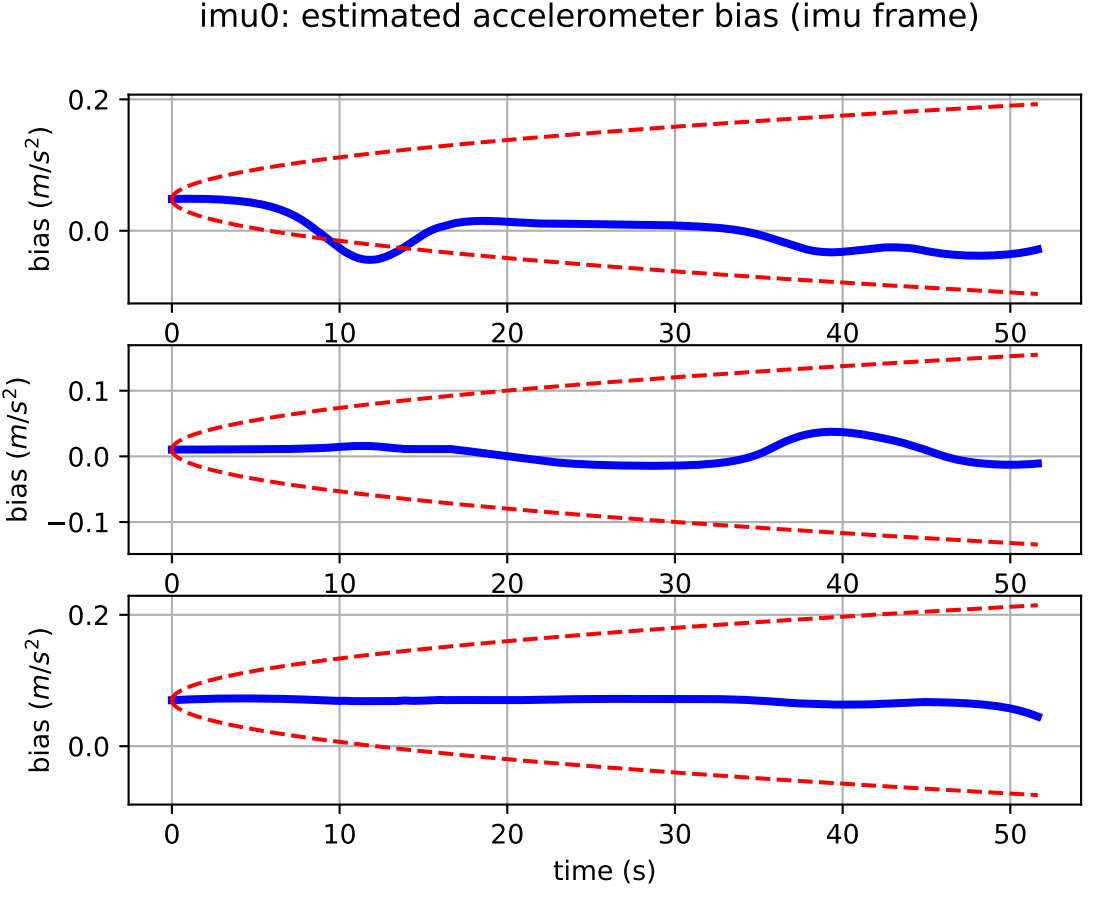
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

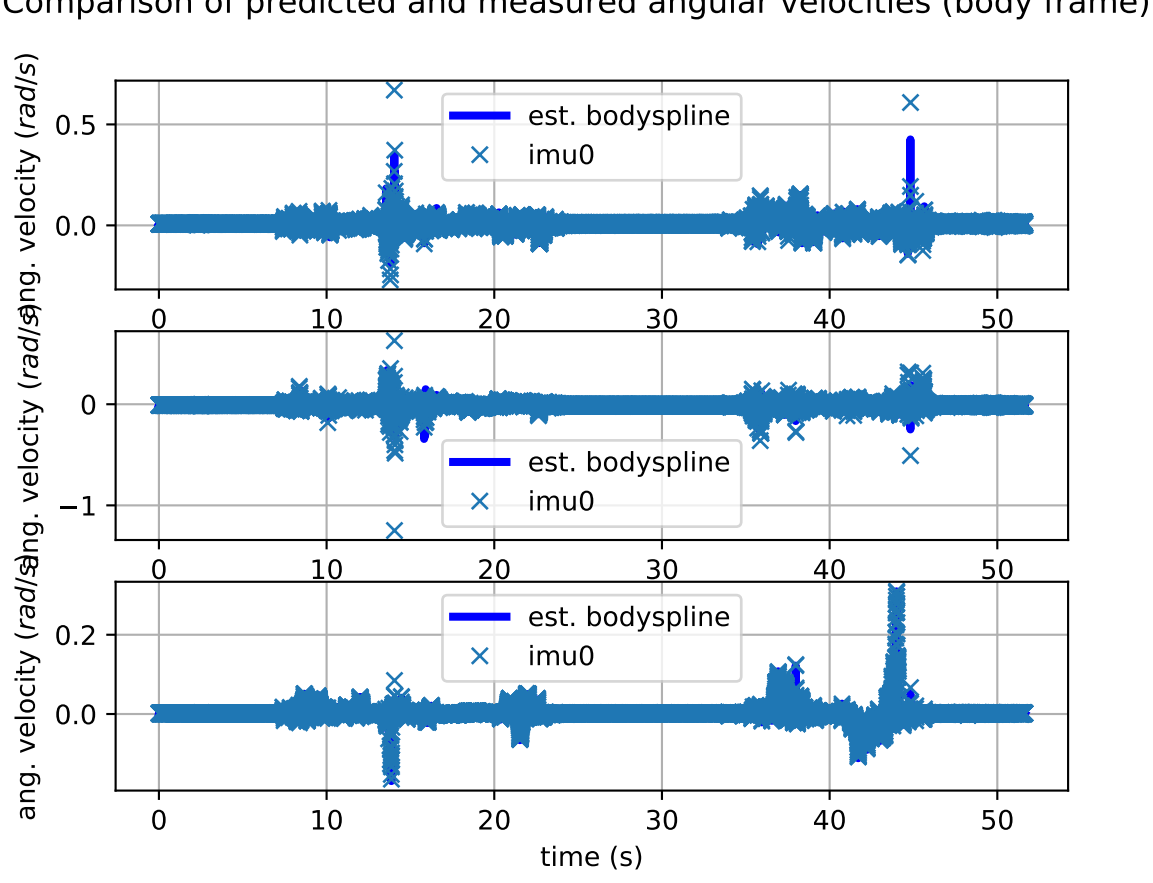

imu0: estimated poses
0.50
0.45
0.40
0.35
0.30
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.02
0.00
0.02
0.04
0.06
0.08
0.10

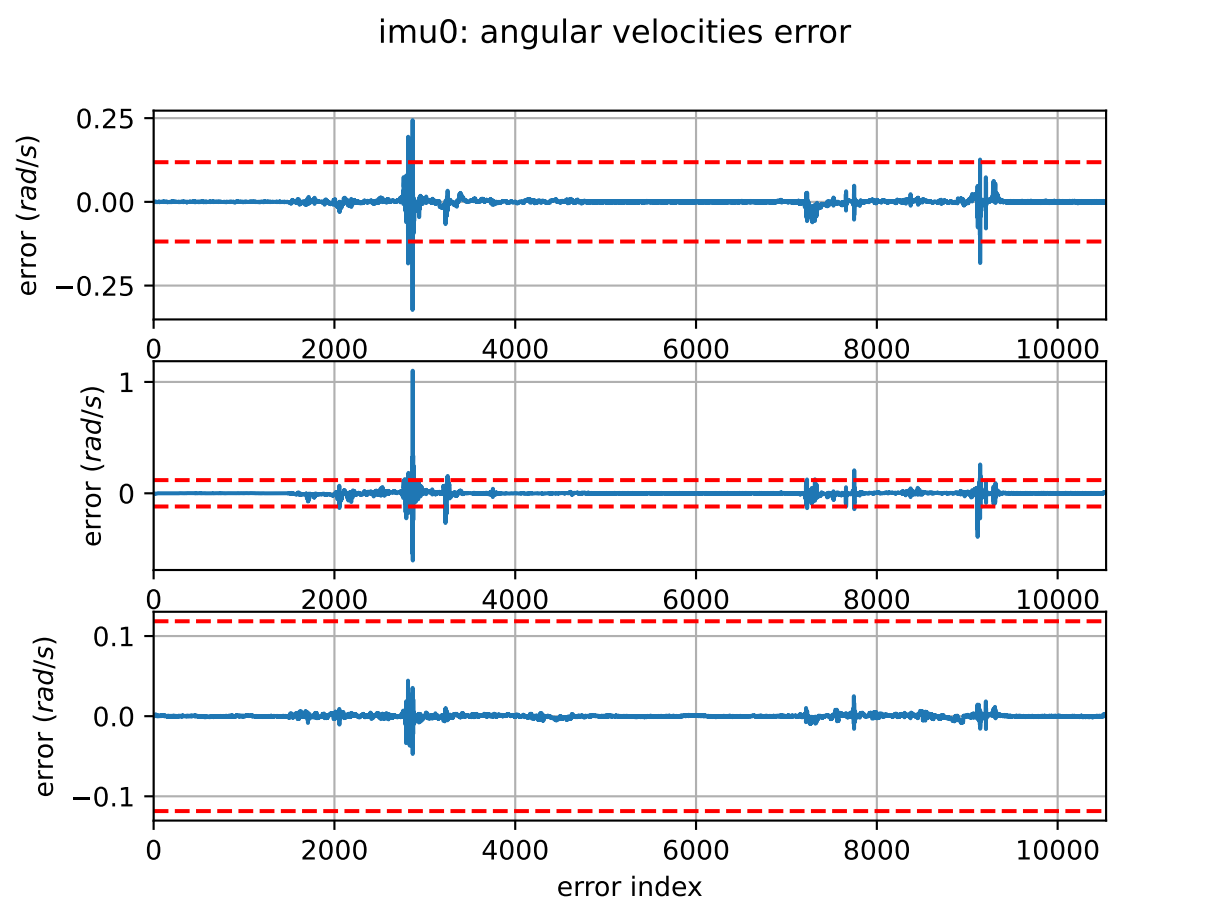

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0002
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

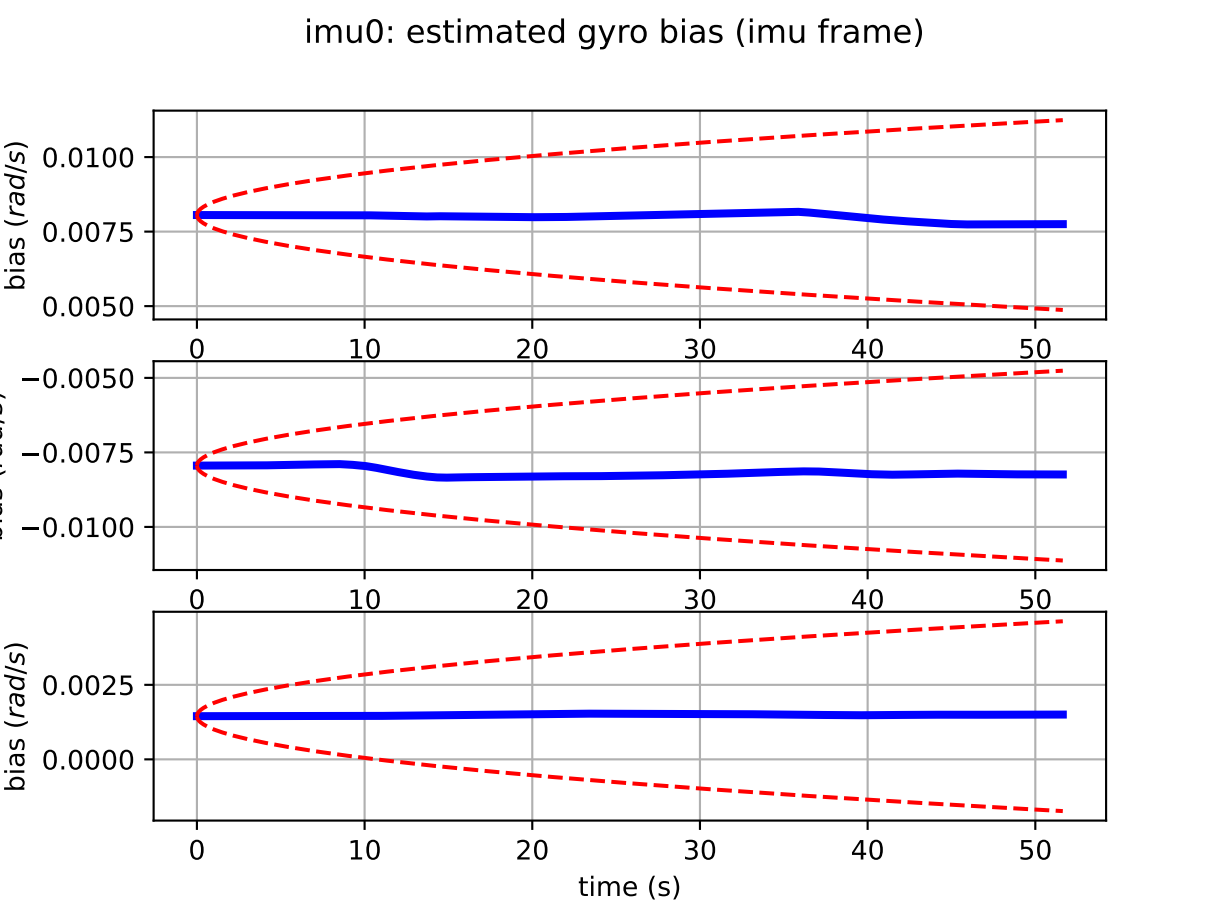

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
2.5
0.0
−2.5
0
10
20
30
40
50
specific force (m/s^2)
est. bodyspline
imu0
2
0
specific force (m/s^2)
est. bodyspline
imu0
20
10
0
time (s)

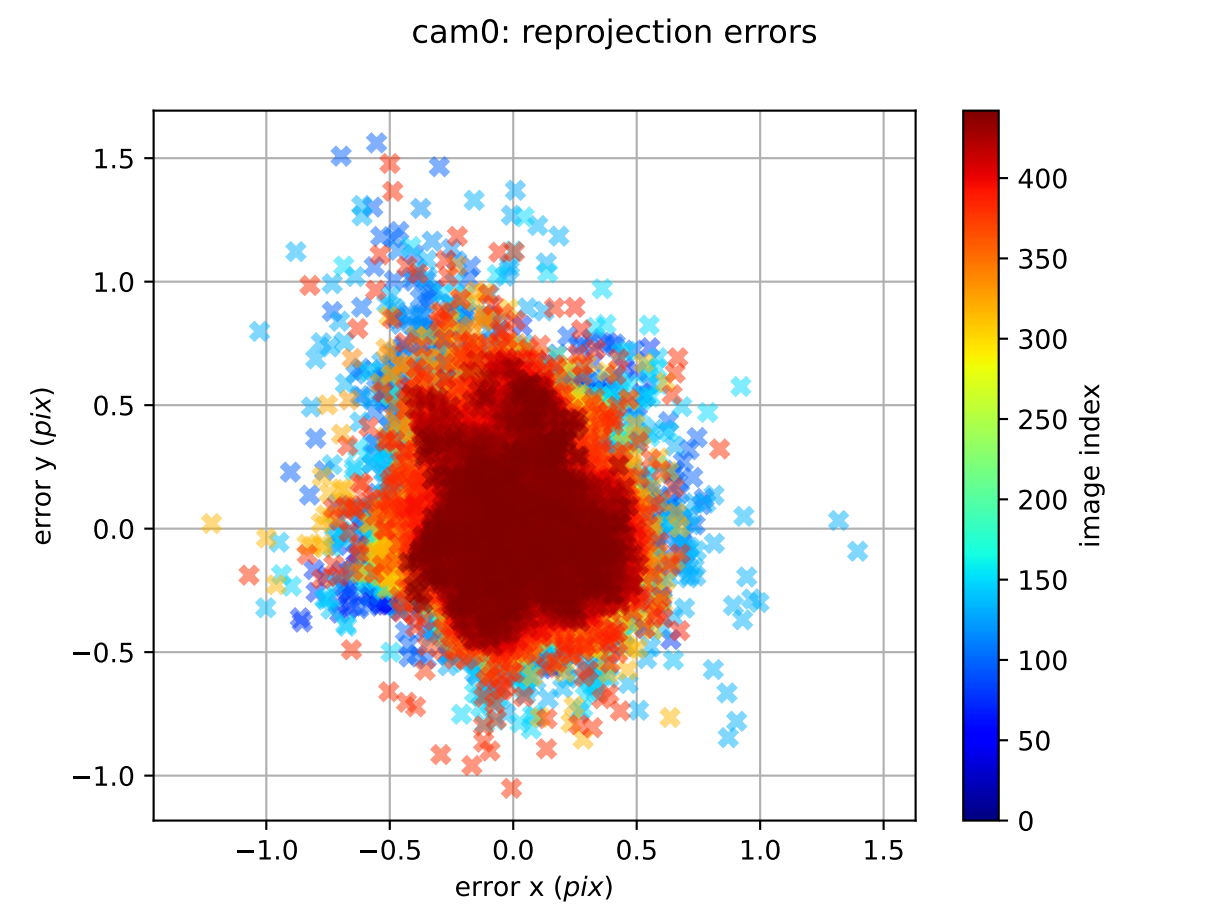

imu0: acceleration error
error (m/s^2)
2
0
−2
0
2000
4000
6000
8000
10000
error (m/s^2)
1
0
−1
0
2000
4000
6000
8000
10000
error (m/s^2)
10
0
−10
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.2
0.0
0
10
20
30
40
50
bias (m/s^2)
0.1
0.0
−0.1
0
10
20
30
40
50
bias (m/s^2)
0.2
0.0
0
10
20
30
40
50
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
0.5
0.0
0
−1
0.2
0.0
0
10
20
30
40
50
time (s)


imu0: angular velocities error
error (rad/s)
0.25
0.00
−0.25
0
2000
4000
6000
8000
10000
error (rad/s)
1
0
0
2000
4000
6000
8000
10000
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.0100
0.0075
0.0050
−0.0050
−0.0075
−0.0100
0.0025
0.0000
0
10
20
30
40
50
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
1.5
1.0
0.5
0.0
−0.5
−1.0
400
350
300
250
200
150
100
50
0